



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.TECH Dual Degree 5Y	IIT Kharagpur	7.40 / 10
2022	MSBSHSE(XII)	Rajarshi Shahu Junior Science College, Latur	82.00%
2020	MSBSHSE(X)	Sri Sri Ravishankar Madhyamik Vidyalaya, Latur	98.20%

INTERNSHIPS

FlyRank AI | Backend AI Engineering Intern Jul '26 - Aug '26

- Architected a **multi-tenant billing system** with a **5-table relational schema**, ensuring isolation and consistency across tenants at scale.
- Built a **scalable async metering pipeline** using **SQLAlchemy** to process usage events into integer-precise cost rollups, preventing drift.
- Eliminated duplicate billing via dual-layer **idempotency** (idempotency keys + **DB constraints**) under concurrent webhook retries scenarios.
- Integrated Stripe-style **resilient webhook processing** with idempotent handlers and retry-safe logic, ensuring **exactly-once processing**.

Trinity | AI Engineering Intern Jun '25 - Aug '25

- Engineered a **production-grade RAG system** using hybrid retrieval (FAISS + BM25) and reranking, achieving **90%+ faithfulness** across 60 queries.
- Improved retrieval performance by boosting **Recall@5** from **~68% to ~91%** via hybrid search and reranking, reducing retrieval noise.
- Optimized **inference efficiency** by reducing context from 20 to 4 chunks via reranking, cutting **LLM cost per query by ~60%** without accuracy loss
- Reduced system latency by **~30% (p95)** using ANN indexing (**HNSW**) and reranker pruning, enabling scalable low-latency retrieval

PROJECTS

CodeAgentPro | Multi-Agent LLM System with RAG, Evaluation & Sandbox Execution | Self Project Mar '26 - Apr '26

Python, FastAPI, LangGraph, LLM (Groq API, Gemini API, Ollama), ChromaDB, Sentence-Transformers, Docker, SSE

- Orchestrated **multi-agent** LangGraph pipeline with adaptive **self-repair** (code, test, debug via RAG memory), enabling autonomous recovery.
- Cut RAG retrieval latency **99.8%** across 3 ChromaDB stores by caching the Sentence-Transformers embedding model as a warmed singleton
- Devised **multi-language eval harness** (Pytest, Node, C++) evaluating Pass@1 across 18-task benchmark via GitHub Actions CI pipelines.
- Unified **3 LLM providers** (Groq, Gemini, Ollama) behind one async client with **prefix routing**, 429 backoff, and **Docker-sandboxed** code execution

GitHub Codebase RAG | Stateful Agentic Code Search Engine | Self Project Dec '25 - Jan '26

LangGraph, FastAPI, PostgreSQL, ChromaDB, Tree-Sitter, DeepEval, Sentence-Transformers, Docker, LangSmith

- Devised a scalable, production-grade **LangGraph RAG pipeline** with AST parsing, **hybrid search** and reranking, boosting **top-5 precision by 35%**.
- Curbed **hallucinations to <3%** using CRAG rewriting and iterative Self-RAG loops, cutting **end-to-end latency by 40%** via fast-mode execution.
- Instituted a robust **DeepEval pipeline** tracking faithfulness, relevance, and latency, achieving **92%+ grounding** across 25+ diverse test cases.

COMPETITION/CONFERENCE

General Championship Data Analytics 2024 | IIT Kharagpur & Evva Health Feb '24 - Mar '24

- Secured **Gold** in the General Championship Data Analytics with a hall team, outperforming 15+ top-performing teams at IIT Kharagpur
- Curated 3 years of SDOH datasets across Census tract, ZIP, and county levels, merging them via crosswalk mapping into one dataset
- Mapped SDOH features to **11 risk categories** using **Sentence-BERT** and **cosine similarity**, then clustered variables within risk using **K-Means**
- Built a percentile-based thresholding algorithm and deployed a Streamlit Risk Calculator app recommending personalized health support

AWARDS AND ACHIEVEMENTS

- Achieved **Knight** on LeetCode (**1891**, nikhilmahi), showcasing strong problem-solving and consistent performance in coding contests.
- Secured Rank **132** and **173** in CodeChef Starters 250 and 251 (nikhilmahi), demonstrating high performance in global contests.
- Attained **3 star** CodeChef (peak **1662**, nikhilmahi) and solved 1500+ DSA problems across platforms, building strong algorithms expertise.

SKILLS AND EXPERTISE

Programming Languages : Python, C++, SQL, MATLAB, JavaScript **ML/AI** : Generative AI, Agentic AI, RAG, Machine Learning, Deep Learning, NLP
Frameworks and Libraries : FastAPI, LangChain, LangGraph, Pandas, NumPy, Scikit-learn, TensorFlow, XGBoost, DeepEval, ChromaDB, SQLite
Tools : Docker, Docker Compose, Git, GitHub, LangSmith, MCP Protocol, n8n, VS Code, Jupyter, Google Colab, Claude Code

COURSEWORK INFORMATION

Coursework : Linear Algebra | Numerical and Complex Analysis | Advanced Calculus | Programming and Data Structures* | Economics
MOOCs : Machine Learning | Deep Learning | Generative AI | Agentic AI | Probability and Statistics | DBMS (*courses with lab)

EXTRA CURRICULAR ACTIVITIES

- Led LLR Hall as **General Secretary** to podium finishes in GC Sports and Games 2025 with **Silver** in Chess and **Bronze** in Water Polo and Tennis.
- Represented the Hall in the General Championship and secured **Gold in Water Polo** (2024) and achieved **4th place in Cricket** (2026).
- Taught village school students weekly as an **NSS volunteer** and led impactful activities that **improved engagement** and learning outcomes.